



SPENDWISE - TECHNOLOGY STACK

Version: 1.0.0 | **Date:** March 7, 2026 | **Status:** ✔ Implemented

1. BACKEND TECHNOLOGIES (CURRENT IMPLEMENTATION)

Core Stack

Technology	Version	Purpose	Implementation
Node.js	v20.20.1	Runtime environment	<code>node server.js</code>
Express.js	^4.18.2	Web framework	<code>const app = express()</code>
MongoDB	^7.5.0	Database	<code>mongoose.connect()</code>
Mongoose	^7.5.0	ODM (Object Data Modeling)	Schema definition

Authentication & Security

Technology	Version	Purpose	Implementation
bcryptjs	^2.4.3	Password hashing	<code>bcrypt.hash(password, 10)</code>
jsonwebtoken	^9.0.2	JWT tokens	<code>jwt.sign({ id }, JWT_SECRET)</code>
dotenv	^16.3.1	Environment variables	<code>require('dotenv').config()</code>
cors	^2.8.5	CORS handling	<code>app.use(cors())</code>

2. ENVIRONMENT CONFIGURATION

.env File

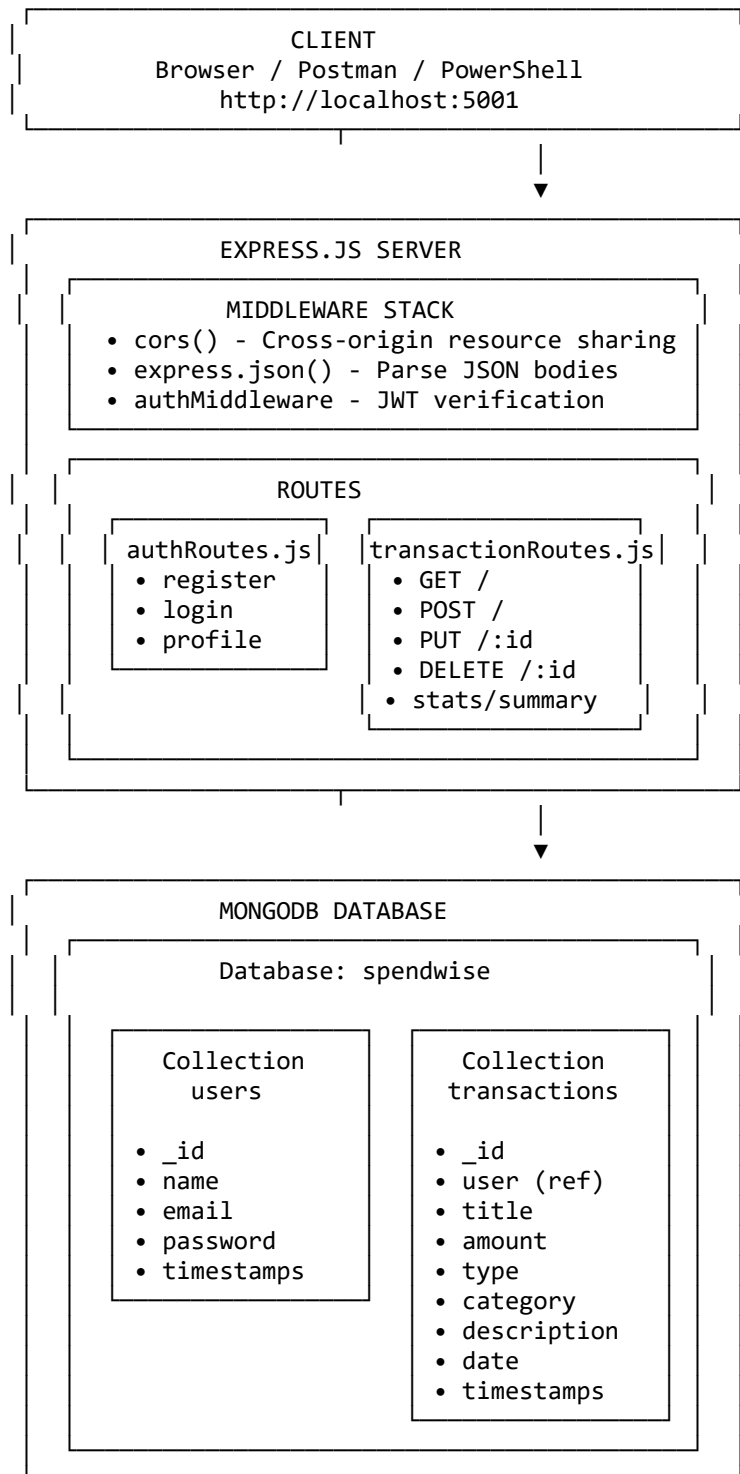
```
PORT=5001 MONGO_URI=mongodb://127.0.0.1:27017/spendwise
JWT_SECRET=spendwise_secret_key
```

package.json Dependencies

```
{ "name": "spendwise-backend", "version": "1.0.0", "main":
"server.js", "scripts": { "start": "node server.js", "dev":
```

```
"nodemon server.js" }, "dependencies": { "express": "^4.18.2",
"mongoose": "^7.5.0", "dotenv": "^16.3.1", "bcryptjs":
"^2.4.3", "jsonwebtoken": "^9.0.2", "cors": "^2.8.5" },
"devDependencies": { "nodemon": "^3.0.1" } }
```

3. ARCHITECTURE DIAGRAM



4. KEY CODE IMPLEMENTATIONS

Server Initialization (server.js)

```
const express = require('express'); const mongoose =
require('mongoose'); const dotenv = require('dotenv'); const
cors = require('cors'); dotenv.config(); const authRoutes =
require('./routes/authRoutes'); const transactionRoutes =
require('./routes/transactionRoutes'); const app = express();
// Middleware app.use(cors()); app.use(express.json()); //
MongoDB Connection mongoose.connect(process.env.MONGO_URI)
.then(() => console.log('✅ MongoDB Connected')) .catch(err =>
console.error('❌ Connection error:', err)); // Routes
app.use('/api/auth', authRoutes); app.use('/api/transactions',
transactionRoutes); app.get('/', (req, res) => { res.json({
message: 'Spendwise API is running', endpoints: { register:
'POST /api/auth/register', login: 'POST /api/auth/login',
profile: 'GET /api/auth/profile', transactions: 'GET
/api/transactions', createTransaction: 'POST
/api/transactions', stats: 'GET
/api/transactions/stats/summary' } }); }); const PORT =
process.env.PORT || 5001; app.listen(PORT, () => {
console.log(`🚀 Server running on port ${PORT}`); });
```

Authentication Middleware (authMiddleware.js)

```
const jwt = require('jsonwebtoken'); const User =
require('../models/User'); const protect = async (req, res,
next) => { let token; if
(req.headers.authorization?.startsWith('Bearer')) { try { token
= req.headers.authorization.split(' ')[1]; const decoded =
jwt.verify(token, process.env.JWT_SECRET); req.user = await
User.findById(decoded.id).select('-password'); next(); } catch
(error) { return res.status(401).json({ message: 'Not
authorized' }); } } if (!token) { return res.status(401).json({
message: 'Not authorized, no token' }); } }; module.exports = {
protect };
```

5. DEVELOPMENT TOOLS

Tool	Purpose	Usage
nodemon	Auto-restart server during development	npm run dev
Postman	API testing	Manual endpoint testing
PowerShell	Test scripts	.\test-spendwise.ps1
MongoDB Compass	Database visualization	View collections
Git	Version control	Source code management

6. TEST SCRIPTS

PowerShell Test Script (test-spendwise.ps1)

```
$apiUrl = "http://localhost:5001" # Test server connection
$response = Invoke-RestMethod -Uri "$apiUrl/" -Method Get
Write-Host "✅ Server: $($response.message)" # Test login
$loginBody = @{"email="test@email.com"; password="123456"} |
ConvertTo-Json $login = Invoke-RestMethod -Uri
"$apiUrl/api/auth/login" -Method Post -Headers @{"Content-
Type"="application/json"} -Body $loginBody $token =
$login.token # Test profile with token $profile = Invoke-
RestMethod -Uri "$apiUrl/api/auth/profile" -Headers
@{"Authorization" = "Bearer $token"}
```

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